Technical Whitepaper

Troubleshooting Battery Run Time Issues on HP Devices

Best Practices for Software Updates, Setting Adjustments, and Performance Tests to Address Battery Run Time Concerns



October 2024

Software Updates for Optimal Performance

A device's battery health is critical to maintaining a healthy overall system. When investigating concerns regarding battery run time between charges, it is imperative to understand the troubleshooting steps to keep an HP device running at the best possible performance while maximizing battery health.

Below are the best practices and recommended settings that ensure healthy battery run time between charges, as well as a series of optional tests that can be run to analyze what might be causing sub-optimal battery health.

Keeping the BIOS and Driver Updated

HP devices that are not running with up-to-date drivers or BIOS can lead to unoptimized battery performance. Keeping these up to date is not only recommended, but critical to maximizing battery performance and maintaining an overall healthy device. For instructions on how to find and download driver or BIOS updates, click the following link: HP Image Assistant | HP Client Management Solutions.

Maintaing an Updated Operating System (OS)

Keeping the operating system (OS) updated is essential for maintaining optimal device performance, including battery health. OS updates often include important bug fixes, security patches, and improvements that can enhance the efficiency of power usage. By regularly updating the OS, users ensure that their device operates smoothly and benefits from the latest performance enhancements, reducing the chances of battery drain caused by outdated software.

Selecting the Right Setting Preference

Utilizing HP Optimized Windows Power Setting

The HP Optimized Power Plan can be utilized to reduce the power a device uses without sacrificing performance. This plan is specifically created with settings tailored to maximize battery health and performance on HP devices. This plan is selected on HP devices by default.

To check that this setting is enabled, follow the below instructions:

1. Open the Windows Start menu and search for Power Options.
2. Under Selected Plan, the currently enabled Power Plan will be displayed.
    a. If HP Optimized is not selected, click Create a Power Plan on the left-hand side menu.
    b. Click the HP Optimized Power Plan and hit Next.
    c. Leave all Turn Off the Display and Put the Computer to Sleep times as is and click Create.

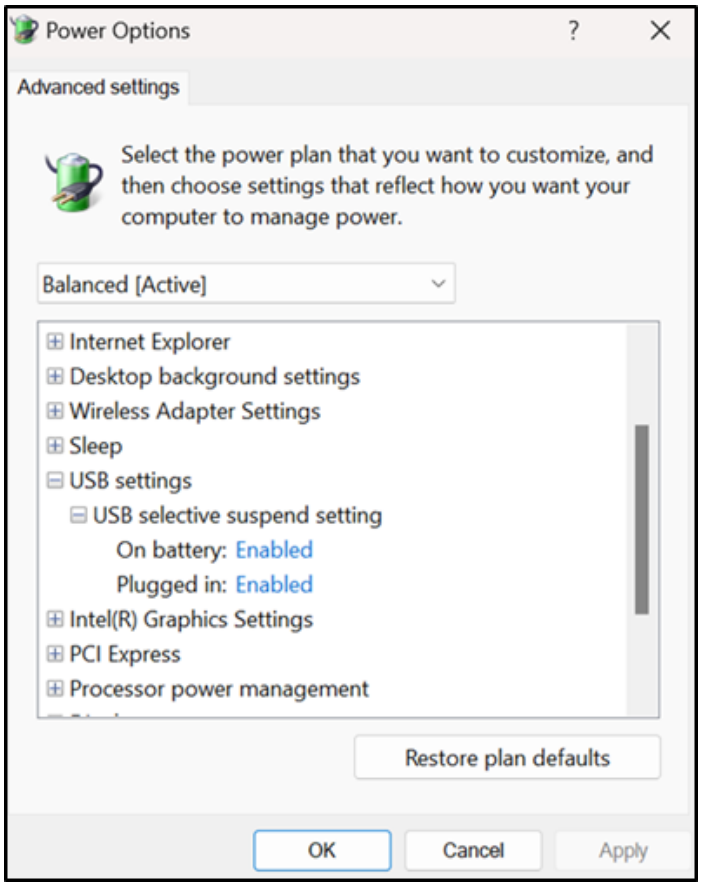
If HP Optimized does not appear as an option on the Create a Power Plan, the device must be updated to the most up-to-date HP BIOS. To do this, refer back to the above section on Keeping Driver and BIOS Updated.

Enabling USB Selective Suspend Setting

Selective Suspend is a Windows OS setting that allows users to put USB ports into a suspended state and builds off what can be done with the Power Saver plan. Similar to how Sleep Mode functions, this setting grants the user the ability to suspend power output to idle devices plugged into the USB ports. To enable this setting, follow the below instructions:

1. Open the Windows Start menu, search for Choose a Power Plan, and open it.
2. In the Choose or Customize a Power Plan panel, select the Change Plan Settings.
3. Click on the Change Advanced Power Settings to open a new settings menu.
4. From the Advanced Power Settings menu, scroll down the list of settings, find the USB Settings option, and expand this option.
5. Expand the USB Selective Suspend Setting to show two enabling options, On Battery and Plugged In.
6. Set both options to Enabled and click Apply.

Image of USB Selective Suspend Setting:



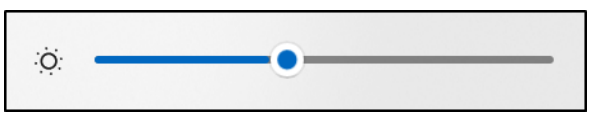
Correct Brightness Setting is Enabled

Utilizing a high brightness setting can greatly affect the rate of battery drain. Keeping the setting at 65% or lower will improve both battery health and battery drain.

To adjust the brightness setting, follow the below instructions:

1. To adjust the brightness setting, follow the below instructions:
2. Find the Brightness Setting slider and adjust it to a comfortable percentage.
3. Click on the Change Advanced Power Settings to open a new settings menu.

Image of Brightening Setting Slider:



Troubleshooting and Performance Tests

Applications / Software Recommendations

Applications or software that are unique to the device's environment could contribute to battery drain. In most instances, these applications utilize more power drain than others, causing the battery power and life to be significantly drained.

A few major examples of these applications / software:

- Security Applications / Software
- Windows Applications / Software
- Mailing Applications / Software
- Messaging Applications / Software

Two common applications that utilize high power consumption are Microsoft Edge and Microsoft Outlook. Both regularly top processor utilization and physical memory processes. While they are both popular applications, it is recommended to close out of them when not in use. This can benefit any power drain that could occur throughout a typical day or even when the device is in a Sleep state.

Another common instance of power drain occurs when multiple Security applications simultaneously scan for malicious software. While beneficial, only one application scan is necessary at a time. Ensuring these applications run interchangeably would promote better power conservation and battery life.

For more information on how to accurately see which applications are affecting the system's power consumption, find the section below on Observing System Performance Using a Task Manager System Test.

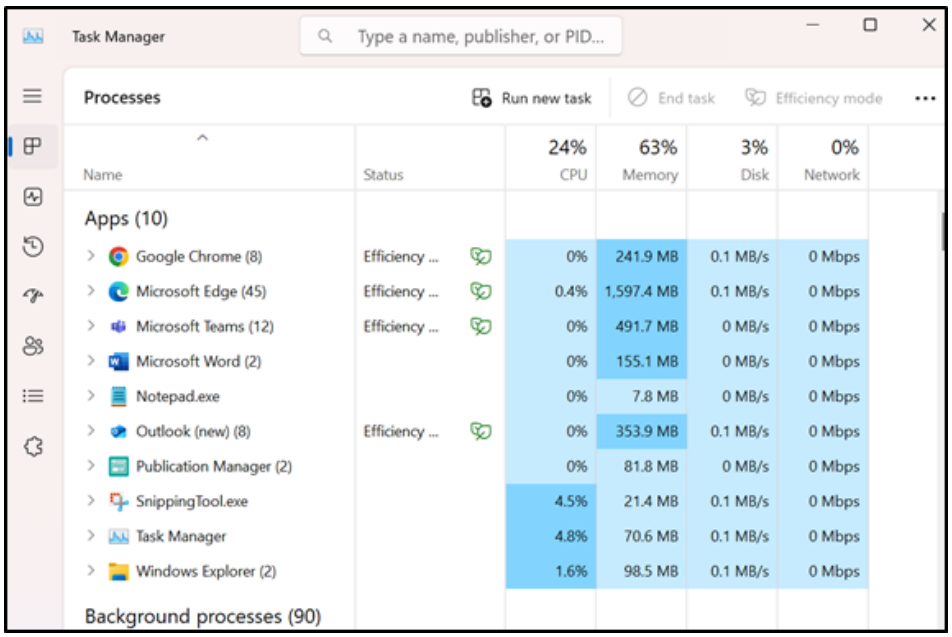
Observing System Performance Using a Task Manager System Test

A device's fan performance is a direct indicator of battery health. When a fan is loud and physically hot, this can lead to the device running ineffectively and can cause the battery to utilize more power than needed. This usually occurs from multiple applications running (either in the background or in general use) and leaving them open even when not in use.

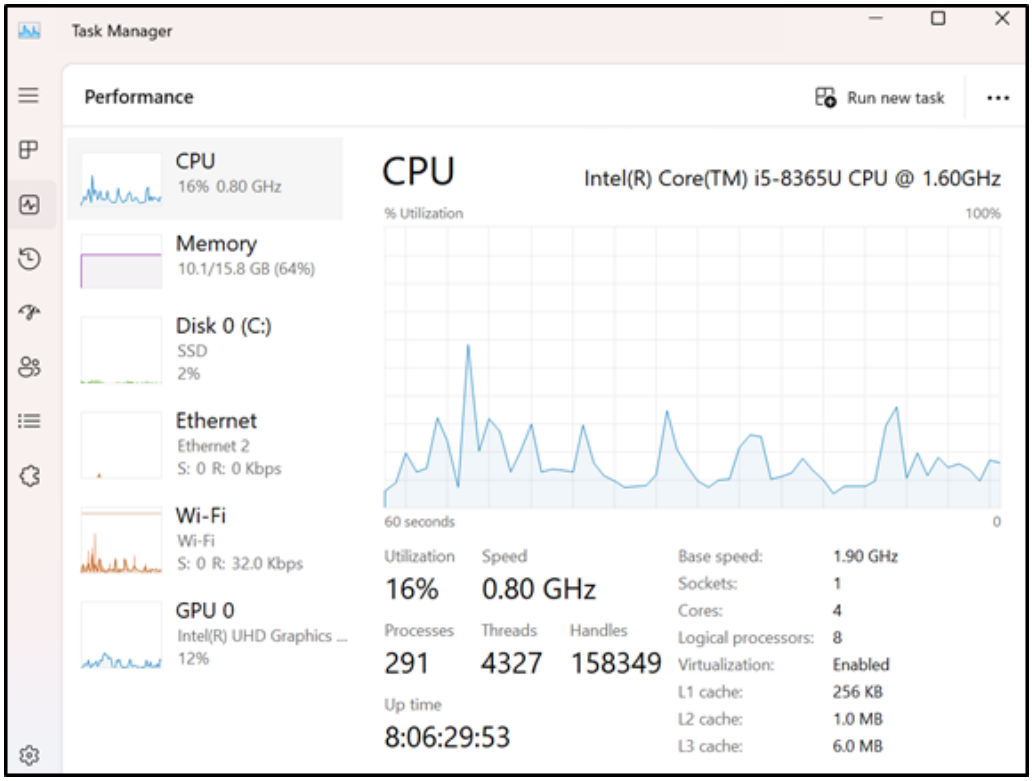
Newer HP models such as the HP EliteBook 1040 G11 and HP Elite x360 1040 G11 offer AI-based battery life innovations in MotionAI and Intelligent Hibernate. Both innovations monitor device usage and PC contextual awareness to maximize the depth of sleep that the device enters when not in use and awakes itself during high usage times. This promotes an overall increase in battery life by up to 15%.

It is beneficial to sporadically check on the system's performance to see a summary of CPU, memory, disk, and network usage. While this does not directly show fan performance, it will provide insight into the overall health of the device and can lead to areas where improvements could be made (for example, what applications are having the largest effect on the system and if they need to be closed out of). To see this system test, follow the following steps:

1. Open the Windows Start menu, search for Task Manager, and open it.
    a. The Task Manager will open the Processes tab, an overview of the overall system's curren performance.
    b. This breakdown will be split into sections such as Apps, Background Processes, or Windows Processes.
    c. If one of the sections shows an unused Task, the option is available to end the Task, preventing it from affecting the system's performance any further.



2. Click on the Performance tab to bring up a detailed breakdown summary of CPU, memory, disk, and network usage within the system.

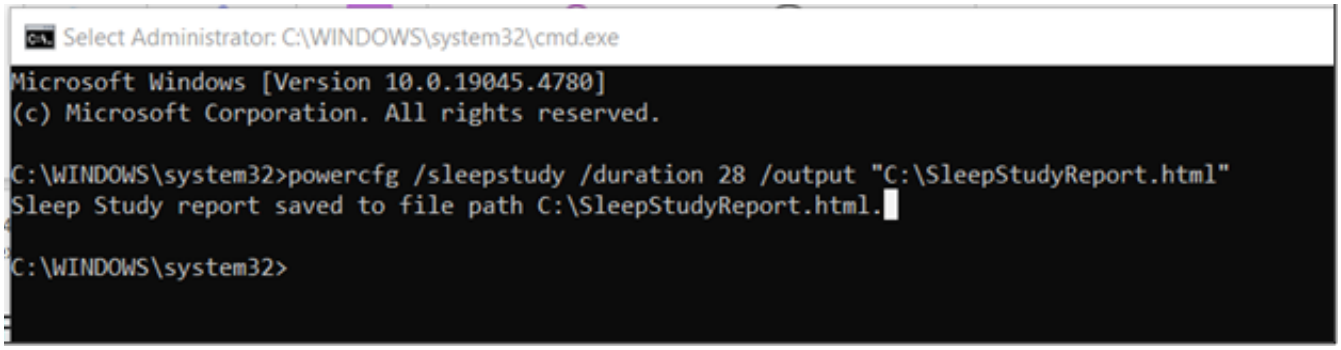


3. Both the Processes and Performance tabs can be used in tandem to see which Tasks are affecting the system's performance. Refer back to the above section on Applications / Software Recommendations for more details on how Tasks unique to the system's environment can affect battery performance.

Running and Reading a Device Sleep Study

A device Sleep Study is a test that can be run to observe what applications are draining the most power while the device is in Sleep Mode. Sleep Studys can be critical to determine which applications can be turned off once the device is in use. This tet is run from the device's Command Prompt console and presents an overall breakdown of every application running during Sleep Mode as well as the impact each has. To run this, follow the below instructions:

1. Open the Windows Start menu, search for Command Prompt, and open it.
2. On the first line, copy and paste in the following command:
   a. powercfg /sleepstudy /duration 28 /output "C:\SleepStudyReport.html"
   Sleep Study report saved to file path C:\SleepStudyReport.html
   b. Please note, to run this command, administrator privileges are required. Ensure the Command Prompt console is being run in Administrator Mode.
3. Once completed, open the report saved to C:\Windows\System32\SleepStudyReport.html.



Once successfully run and opened, the report will display the applications draining the most power in red and the percentage of how much battery is being used. Clicking on the specific row will show what the application is, how much battery is being drained, and at what time the power is being consumed.

Image of Sleep Study Introduction:

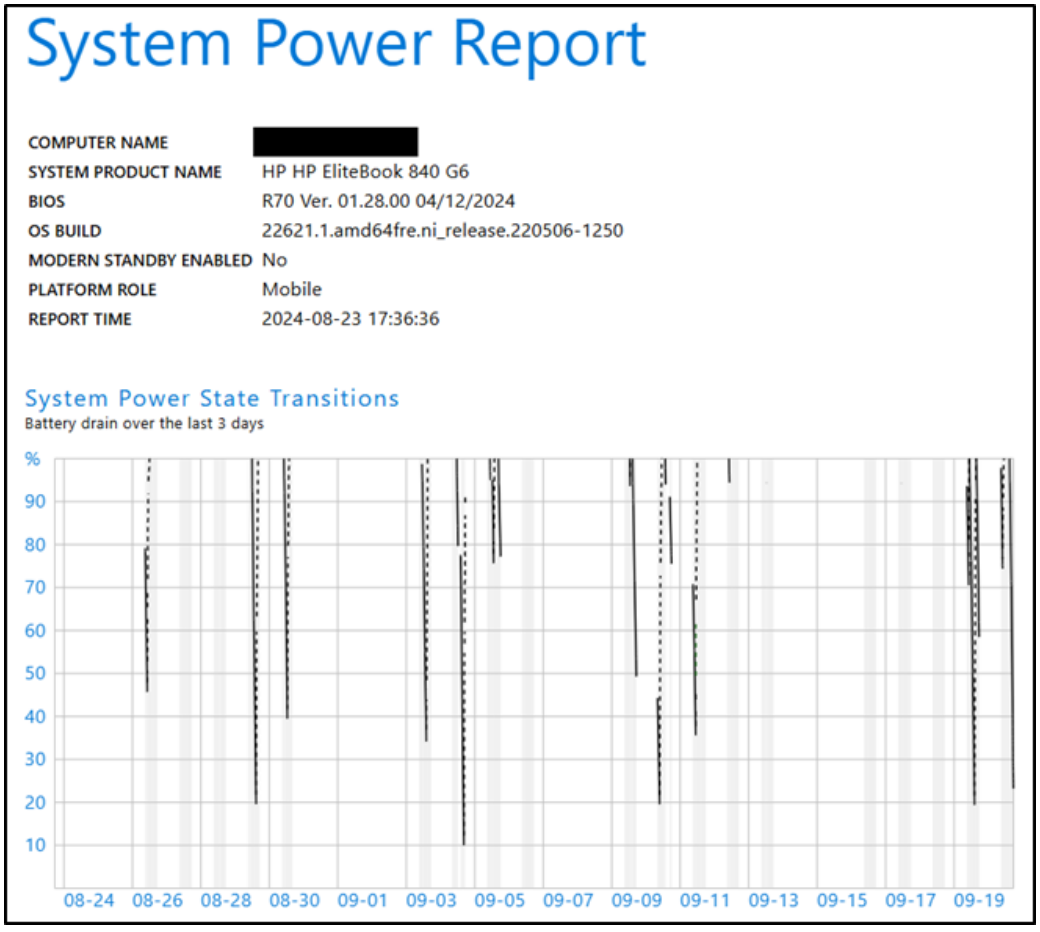
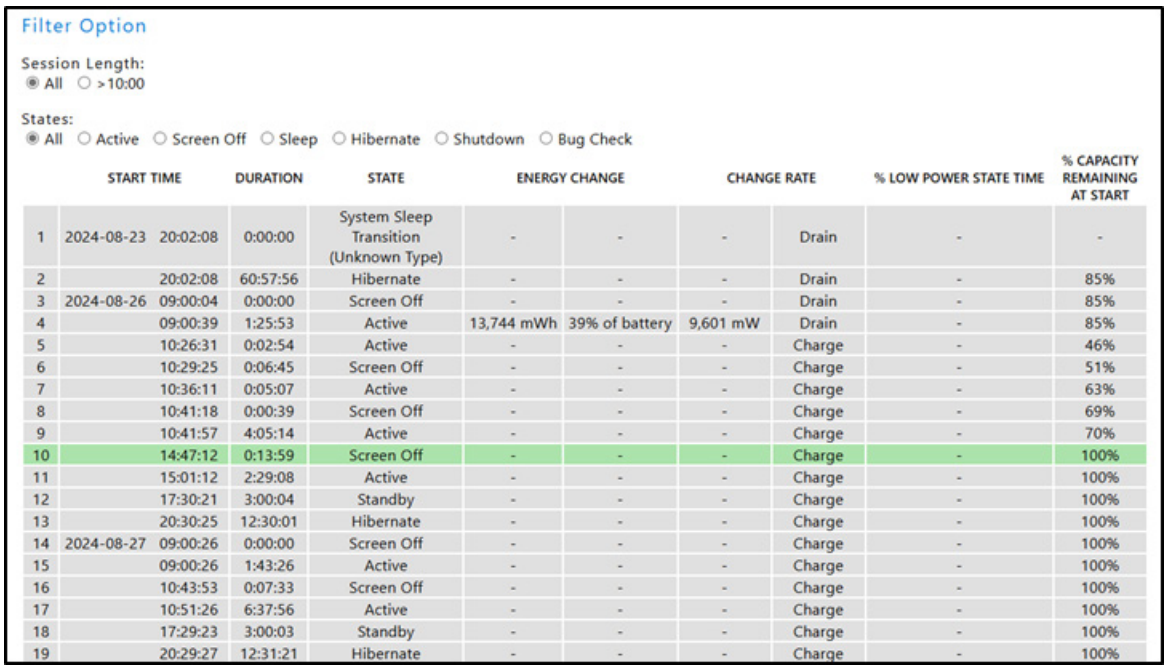


Image of Sleep Study Chart:



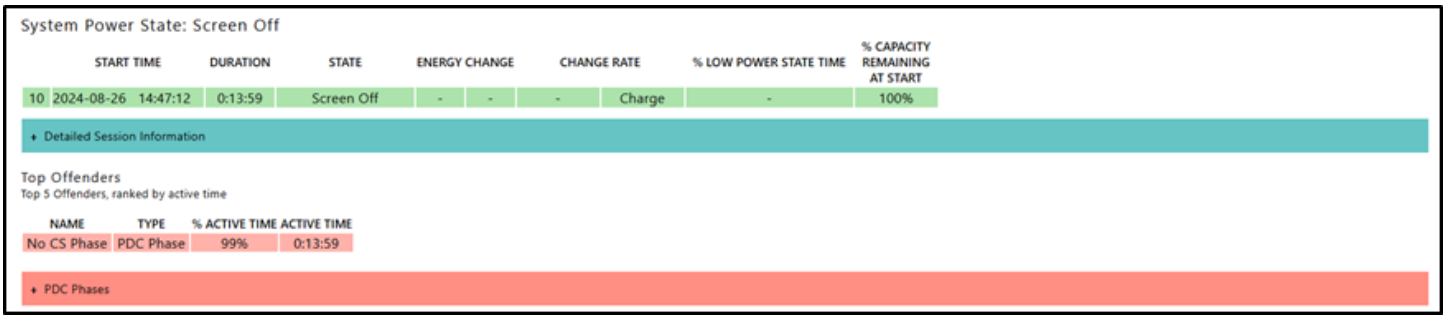
Filter Option

Session Length:
◉ All ○ >10:00

States:
◉ All ○ Active ○ Screen Off ○ Sleep ○ Hibernate ○ Shutdown ○ Bug Check

	START TIME	DURATION	STATE	ENERGY CHANGE		CHANGE RATE		% LOW POWER STATE TIME	% CAPACITY REMAINING AT START
1	2024-08-23 20:02:08	0:00:00	System Sleep Transition (Unknown Type)	-	-	-	Drain	-	-
2	20:02:08	60:57:56	Hibernate	-	-	-	Drain	-	85%
3	2024-08-26 09:00:04	0:00:00	Screen Off	-	-	-	Drain	-	85%
4	09:00:39	1:25:53	Active	13,744 mWh	39% of battery	9,601 mW	Drain	-	85%
5	10:26:31	0:02:54	Active	-	-	-	Charge	-	46%
6	10:29:25	0:06:45	Screen Off	-	-	-	Charge	-	51%
7	10:36:11	0:05:07	Active	-	-	-	Charge	-	63%
8	10:41:18	0:00:39	Screen Off	-	-	-	Charge	-	69%
9	10:41:57	4:05:14	Active	-	-	-	Charge	-	70%
10	14:47:12	0:13:59	Screen Off	-	-	-	Charge	-	100%
11	15:01:12	2:29:08	Active	-	-	-	Charge	-	100%
12	17:30:21	3:00:04	Standby	-	-	-	Charge	-	100%
13	20:30:25	12:30:01	Hibernate	-	-	-	Charge	-	100%
14	2024-08-27 09:00:26	0:00:00	Screen Off	-	-	-	Charge	-	100%
15	09:00:26	1:43:26	Active	-	-	-	Charge	-	100%
16	10:43:53	0:07:33	Screen Off	-	-	-	Charge	-	100%
17	10:51:26	6:37:56	Active	-	-	-	Charge	-	100%
18	17:29:23	3:00:03	Standby	-	-	-	Charge	-	100%
19	20:29:27	12:31:21	Hibernate	-	-	-	Charge	-	100%

Image of Sleep Test Detailed Application Information:



System Power State: Screen Off

	START TIME	DURATION	STATE	ENERGY CHANGE		CHANGE RATE		% LOW POWER STATE TIME	% CAPACITY REMAINING AT START
10	2024-08-26 14:47:12	0:13:59	Screen Off	-	-	-	Charge	-	100%

▸ Detailed Session Information

Top Offenders
Top 5 Offenders, ranked by active time

NAME	TYPE	% ACTIVE TIME	ACTIVE TIME
No CS Phase	PDC Phase	99%	0:13:59

▸ PDC Phases

Related Articles:
For more information on Battery Health and ways to improve battery performance, see the following links: